

Detailed Component-by-Component Specification

Page: Principal Timetable Page

Module: School Timetable

Product: NASSAQ

1) Root Page Component

Component Name

PrincipalTimetablePage

Responsibility

للمصفحة بالكامل root container هذا هو ال
مسؤول عن:

- تحميل بيانات الصفحة
- إدارة الحالة العامة
- تنسيق الأقسام الرئيسية
- تمرير البيانات إلى المكونات الفرعية
- والدوال الرئيسية modals إدارة ال

Props

```
type PrincipalTimetablePageProps = {  
  
  schoolId: string  
  
  academicYearId?: string  
  
  termId?: string  
  
  principalUserId: string  
  
}
```

State

```
type PrincipalTimetablePageState = {  
  pageStatus: "loading" | "ready" | "error"  
  generationStatus: "idle" | "running" | "success" | "failed"  
  activeViewMode: "class" | "teacher" | "grade" | "day"  
  selectedVersionId: string | null  
  selectedClassId: string | null  
  selectedTeacherId: string | null  
  selectedGradeId: string | null  
  selectedWeekday: number | null  
  selectedSubjectId: string | null  
  readinessSummary: ReadinessSummary | null  
  timetableVersionSummary: TimetableVersionSummary | null  
  timetableGridData: TimetableGridData | null  
  diagnosticsSummary: DiagnosticsSummary | null  
  issuesSummary: IssueItem[]  
  insightsSummary: TimetableInsights | null  
  modalState: {  
    generationModalOpen: boolean  
    partialRegenerationModalOpen: boolean  
    publishModalOpen: boolean  
    archiveModalOpen: boolean  
    diagnosticsModalOpen: boolean  
    sessionDetailsOpen: boolean  
  }  
  selectedSessionEntryId: string | null  
  toastMessage: ToastPayload | null  
  errorMessage: string | null
```

```
}
```

Actions

- `loadPageData()`
- `refreshAllData()`
- `openGenerationModal()`
- `openPartialRegenerationModal()`
- `openPublishModal()`
- `openArchiveModal()`
- `openDiagnosticsModal()`
- `openSessionDetails(entryId)`
- `closeAllModals()`
- `handleGenerationCompleted()`
- `handleVersionChanged(versionId)`
- `handleViewModeChanged(mode)`

Child Components

- `TimetablePageHeader`
- `TimetableStatusBanner`
- `TimetableActionBar`
- `TimetableReadinessPanel`
- `TimetableViewControls`
- `TimetableGridSection`
- `TimetableInsightsPanel`
- `TimetableVersionManager`
- `TimetableIssuesSection`
- all related modals

2) Page Header Component

Component Name

`TimetablePageHeader`

Responsibility

عرض عنوان الصفحة وسياق المدرسة الحالي.

Props

```
type TimetablePageHeaderProps = {
```

```
  schoolNameAr: string
```

```
  academicYearName: string
```

```
termName: string

totalClasses: number

totalTeachers: number

totalTeachingSlots: number

currentVersionStatus: "none" | "draft" | "published" | "generating"

}
```

State

داخلي معقد state لا يحتاج

Actions

rendering، رئيسية، فقط actions لا توجد

UI Elements

- page title
- subtitle
- breadcrumb
- metadata chips

الحالات

- normal
- loading skeleton

3) Status Banner Component

Component Name

TimetableStatusBanner

Responsibility

عرض الحالة التشغيلية الحالية للجدول

Props

```
type TimetableStatusBannerProps = {

  status: "no_timetable" | "draft_exists" | "published_exists" | "generating" | "failed"
```

```
versionName?: string
generatedAt?: string
publishedAt?: string
generatedBy?: string
qualityScore?: number
warningsCount?: number
conflictsCount?: number
message?: string
onViewDiagnostics?: () => void
onPublishDraft?: () => void
onStartGeneration?: () => void
}
```

State

stateless. يمكن أن يكون

Actions

- trigger diagnostics
- trigger publish
- trigger generation

الحالات

- Empty info banner
- Draft info banner
- Published success banner
- Processing banner
- Failure banner

4) Action Bar Component

Component Name

TimetableActionBar

Responsibility

عرض جميع الإجراءات الأساسية الخاصة بالجدول.

Props

```
type TimetableActionBarProps = {  
  
  canGenerate: boolean  
  
  canPublish: boolean  
  
  canArchive: boolean  
  
  canPartialRegenerate: boolean  
  
  isGenerating: boolean  
  
  hasDraftVersion: boolean  
  
  hasPublishedVersion: boolean  
  
  onGenerateClick: () => void  
  
  onPartialRegenerateClick: () => void  
  
  onPublishClick: () => void  
  
  onArchiveClick: () => void  
  
  onDiagnosticsClick: () => void  
  
  onGoToSettingsClick: () => void  
  
}
```

State

```
type TimetableActionBarState = {  
  
  hoveredAction: string | null  
  
}
```

Actions

- open generation modal
- open partial regeneration modal
- open publish modal
- open archive modal
- open diagnostics
- navigate to school settings

UI Elements

- Primary AI button
- secondary buttons
- tooltips on disabled buttons

5) Primary AI Button Component

Component Name

AITimetableGenerateButton

Responsibility

زر الإجراء الرئيسي في الصفحة

Props

```
type AITimetableGenerateButtonProps = {  
  
  disabled: boolean  
  
  loading: boolean  
  
  readinessPassed: boolean  
  
  onClick: () => void  
  
  disabledReason?: string  
  
}
```

State

مهم state لا يوجد.

Actions

- click → open generation flow

حالات الزر

- enabled
- disabled
- loading
- success flash after completion optional

6) Readiness Panel Component

Component Name

TimetableReadinessPanel

Responsibility

عرض جاهزية البيانات المطلوبة لتوليد الجدول.

Props

```
type TimetableReadinessPanelProps = {  
  items: ReadinessItem[]  
  overallStatus: "ready" | "partial" | "blocked"  
  onFixItem: (itemKey: string) => void  
  onRefresh: () => void  
}
```

Supporting Types

```
type ReadinessItem = {  
  key: string  
  label: string  
  status: "complete" | "missing" | "warning"  
  description: string  
  fixRoute?: string  
  fixActionLabel?: string  
}
```

State

```
type TimetableReadinessPanelState = {  
  expanded: boolean  
}
```

Actions

- expand/collapse
- click fix item
- refresh readiness

Child Components

- ReadinessSummaryCard
- ReadinessChecklistItem

7) Readiness Checklist Item

Component Name

ReadinessChecklistItem

Responsibility

readiness validation. عرض بند واحد من

Props

```
type ReadinessChecklistItemProps = {  
  
  item: ReadinessItem  
  
  onFix: (itemKey: string) => void  
  
}
```

State

لا يوجد.

Actions

- fix action
- open related settings page

8) View Controls Wrapper

Component Name

TimetableViewControls

Responsibility

التحكم في طريقة عرض الجدول والفلاتر والبحث.

Props

```
type TimetableViewControlsProps = {  
  activeViewMode: "class" | "teacher" | "grade" | "day"  
  selectedClassId: string | null  
  selectedTeacherId: string | null  
  selectedGradeId: string | null  
  selectedWeekday: number | null  
  selectedVersionId: string | null  
  selectedSubjectId: string | null  
  classes: OptionItem[]  
  teachers: OptionItem[]  
  grades: OptionItem[]  
  subjects: OptionItem[]  
  weekdays: OptionItem[]  
  versions: OptionItem[]  
  searchQuery: string  
  toggles: TimetableViewToggles  
  onViewModeChange: (mode: ViewMode) => void  
  onClassChange: (id: string | null) => void  
  onTeacherChange: (id: string | null) => void  
  onGradeChange: (id: string | null) => void  
  onWeekdayChange: (day: number | null) => void  
  onVersionChange: (id: string) => void  
  onSubjectChange: (id: string | null) => void  
  onSearchChange: (value: string) => void
```

```
onToggleChange: (next: Partial<TimetableViewToggles>) => void
}
```

Supporting Types

```
type TimetableViewToggles = {
  showBreaks: boolean
  showPrayer: boolean
  showWarnings: boolean
  showColorCoding: boolean
  showCoreSubjectsOnly: boolean
}
```

State

في الموبايل filter drawer بسيط مثل فتح/إغلاق UI state قد يحتوي فقط على

Child Components

- TimetableViewModeTabs
- TimetableFiltersBar
- TimetableSearchInput
- TimetableToggleControls

9) View Mode Tabs

Component Name

TimetableViewModeTabs

Responsibility

التبديل بين أوضاع عرض الجدول.

Props

```
type TimetableViewModeTabsProps = {
  activeMode: "class" | "teacher" | "grade" | "day"
  onChange: (mode: ViewMode) => void
}
```

```
}
```

Actions

- switch mode

الحالات

- active tab
- inactive tabs

10) Filters Bar

Component Name

TimetableFiltersBar

Responsibility

عرض الفلاتر الديناميكية حسب نوع العرض.

Props

```
type TimetableFiltersBarProps = {
```

```
  mode: ViewMode
```

```
  classes: OptionItem[]
```

```
  teachers: OptionItem[]
```

```
  grades: OptionItem[]
```

```
  subjects: OptionItem[]
```

```
  weekdays: OptionItem[]
```

```
  versions: OptionItem[]
```

```
  selectedClassId: string | null
```

```
  selectedTeacherId: string | null
```

```
  selectedGradeId: string | null
```

```
  selectedSubjectId: string | null
```

```
  selectedWeekday: number | null
```

```
  selectedVersionId: string | null
```

```
onClassChange: (id: string | null) => void
onTeacherChange: (id: string | null) => void
onGradeChange: (id: string | null) => void
onSubjectChange: (id: string | null) => void
onWeekdayChange: (day: number | null) => void
onVersionChange: (id: string) => void
}
```

State

داخلي أساسي logic لا يوجد

Actions

- set filters
- clear filter values if mode changes

11) Search Input Component

Component Name

TimetableSearchInput

Responsibility

البحث النصي داخل الجدول

Props

```
type TimetableSearchInputProps = {
  value: string
  placeholder?: string
  onChange: (value: string) => void
  onClear: () => void
}
```

State

إذا لزم debounced internal state يمكن إضافة.

12) Toggle Controls Component

Component Name

TimetableToggleControls

Responsibility

الخاصة بطريقة إظهار البيانات toggles عرض.

Props

```
type TimetableToggleControlsProps = {  
  toggles: TimetableViewToggles  
  onChange: (next: Partial<TimetableViewToggles>) => void  
}
```

Actions

- toggle on/off

13) Main Grid Section Wrapper

Component Name

TimetableGridSection

Responsibility

احتواء الجدول الفعلي وحالاته المختلفة.

Props

```
type TimetableGridSectionProps = {  
  loading: boolean  
  hasData: boolean  
  gridData: TimetableGridData | null  
  activeViewMode: ViewMode
```

```
onSessionClick: (entryId: string) => void  
}
```

State

معقد state لا يوجد.

Child Components

- TimetableGridHeader
- TimetableGrid
- TimetableEmptyGridState
- TimetableGridSkeleton

14) Timetable Grid Component

Component Name

TimetableGrid

Responsibility

عرض الشبكة الرئيسية للجدول.

Props

```
type TimetableGridProps = {  
  columns: TimetableGridColumn[]  
  rows: TimetableGridRow[]  
  entries: TimetableGridEntry[]  
  showBreaks: boolean  
  showPrayer: boolean  
  showWarnings: boolean  
  showColorCoding: boolean  
  onEntryClick: (entryId: string) => void  
}
```

Supporting Types

```
type TimetableGridColumn = {  
    key: string  
    label: string  
}
```

```
type TimetableGridRow = {  
    key: string  
    label: string  
    startTime: string  
    endTime: string  
    blockType: "period" | "break" | "prayer" | "assembly" | "custom"  
}
```

```
type TimetableGridEntry = {  
    id: string  
    columnKey: string  
    rowKey: string  
    entryType: "lecture" | "break" | "prayer" | "assembly" | "custom" | "empty"  
    subjectName?: string  
    teacherName?: string  
    className?: string  
    roomName?: string  
    isAiGenerated?: boolean  
    isLocked?: boolean  
    hasWarning?: boolean  
    warningCount?: number  
    status?: "scheduled" | "modified" | "locked" | "cancelled"
```

```
}
```

State

hovered cell أو focused cell. يمكن إضافة state ل

Actions

- click on entry
- keyboard navigate optional
- hover reveal quick info

15) Timetable Cell Component

Component Name

TimetableCell

Responsibility

.عرض خلية واحدة داخل الشبكة

Props

```
type TimetableCellProps = {  
  entry: TimetableGridEntry  
  showWarnings: boolean  
  showColorCoding: boolean  
  onClick: (entryId: string) => void  
}
```

State

```
type TimetableCellState = {  
  hovered: boolean  
}
```

Actions

- open session details
- show quick tooltip on hover

Variants

- LectureCell
- BreakCell
- PrayerCell
- EmptyCell

16) Empty Grid State

Component Name

TimetableEmptyGridState

Responsibility

إظهار حالة عدم وجود بيانات داخل مساحة الجدول.

Props

```
type TimetableEmptyGridStateProps = {  
  reason: "no_version" | "no_filter_selection" | "no_matching_results"  
  
  onGenerate?: () => void  
  
  onClearFilters?: () => void  
  
}
```

Actions

- generate
- clear filters

17) Insights Panel Component

Component Name

TimetableInsightsPanel

Responsibility

عرض ملخص ذكي وتحليلات النسخة الحالية.

Props

```
type TimetableInsightsPanelProps = {  
  insights: TimetableInsights | null  
  loading: boolean  
  onOpenDiagnostics: () => void  
}
```

Supporting Types

```
type TimetableInsights = {  
  totalRequiredSessions: number  
  totalAssignedSessions: number  
  totalUnassignedSessions: number  
  qualityScore: number  
  warningsCount: number  
  hardConflictsCount: number  
  affectedTeachersCount: number  
  affectedClassesCount: number  
  topOverloadedClass?: string  
  topOverloadedTeacher?: string  
  topClusteredSubject?: string  
  aiNotes: string[]  
}
```

Child Components

- InsightMetricCard
- AllInsightNotice

18) Insight Metric Card

Component Name

InsightMetricCard

Responsibility

واحدة metric عرض.

Props

```
type InsightMetricCardProps = {  
  label: string  
  
  value: string | number  
  
  tone?: "default" | "success" | "warning" | "danger" | "info"  
  
  helperText?: string  
  
}
```

19) Version Management Component

Component Name

TimetableVersionManager

Responsibility

عرض نسخ الجدول وإدارتها.

Props

```
type TimetableVersionManagerProps = {  
  versions: TimetableVersionCardItem[]  
  
  selectedVersionId: string | null  
  
  onSelectVersion: (id: string) => void  
  
  onPublishVersion: (id: string) => void  
  
  onArchiveVersion: (id: string) => void  
  
  onCompareVersion: (id: string) => void  
  
  onRestoreAsBase: (id: string) => void  
  
}
```

Supporting Types

```
type TimetableVersionCardItem = {  
  
  id: string  
  
  versionName: string  
  
  status: "draft" | "published" | "archived" | "failed"  
  
  generationMode: string  
  
  qualityScore: number  
  
  conflictsCount: number  
  
  warningsCount: number  
  
  generatedAt: string  
  
  generatedBy: string  
  
}
```

Child Components

- TimetableVersionCard

20) Version Card Component

Component Name

TimetableVersionCard

Responsibility

عرض نسخة واحدة من نسخ الجدول.

Props

```
type TimetableVersionCardProps = {  
  
  version: TimetableVersionCardItem  
  
  selected: boolean  
  
  onSelect: () => void  
  
  onPublish: () => void  
  
  onArchive: () => void
```

```
onCompare: () => void  
  
onRestoreAsBase: () => void  
  
}
```

Actions

- select version
- publish
- archive
- compare
- restore as base

21) Issues Section Component

Component Name

TimetableIssuesSection

Responsibility

عرض قائمة المشاكل والتحذيرات المرتبطة بالنسخة الحالية.

Props

```
type TimetableIssuesSectionProps = {  
  
  items: IssueItem[]  
  
  loading: boolean  
  
  onOpenIssue: (issueId: string) => void  
  
  onFixIssue: (issueId: string) => void  
  
}
```

Supporting Types

```
type IssueItem = {  
  
  id: string  
  
  type: string  
  
  severity: "low" | "medium" | "high" | "critical"  
  
  title: string
```

```
description: string

relatedEntityType?: string

relatedEntityName?: string

recommendedAction?: string

}
```

Child Components

- IssueAccordionItem
- IssueSeverityBadge

22) Diagnostics Modal Component

Component Name

TimetableDiagnosticsModal

Responsibility

عرض تفاصيل تشغيل المحرك.

Props

```
type TimetableDiagnosticsModalProps = {

  open: boolean

  diagnostics: DiagnosticsSummary | null

  loading: boolean

  onClose: () => void

}
```

Supporting Types

```
type DiagnosticsSummary = {

  runId: string

  startedAt: string

  finishedAt?: string

  status: "running" | "completed" | "failed"
```

```
totalUnits: number

assignedUnits: number

unassignedUnits: number

hardConflicts: number

softConflicts: number

logs: DiagnosticLogGroup[]

}
```

Child Components

- DiagnosticsMetaPanel
- DiagnosticsLogGroup
- DiagnosticsIssueList

23) Generation Modal Component

Component Name

AlTimetableGenerationModal

Responsibility

بدء عملية التوليد الكامل.

Props

```
type AlTimetableGenerationModalProps = {

  open: boolean

  readinessSummary: ReadinessSummary | null

  generationInputSummary: GenerationInputSummary | null

  submitting: boolean

  onConfirm: (payload: FullGenerationPayload) => void

  onClose: () => void

}
```

Supporting Types

```
type GenerationInputSummary = {  
  
  totalClasses: number  
  
  totalTeachers: number  
  
  totalSubjects: number  
  
  totalTeachingSlots: number  
  
}
```

```
type FullGenerationPayload = {  
  
  generationMode: "full"  
  
  usePublishedAsBaseline?: boolean  
  
}
```

State

```
type AITimetableGenerationModalState = {  
  
  usePublishedAsBaseline: boolean  
  
}
```

Actions

- confirm generation
- close modal

24) Partial Regeneration Modal

Component Name

PartialRegenerationModal

Responsibility

إعادة معالجة جزء محدد من الجدول.

Props

```
type PartialRegenerationModalProps = {  
  
  open: boolean
```

```
classes: OptionItem[]

teachers: OptionItem[]

weekdays: OptionItem[]

subjects: OptionItem[]

submitting: boolean

onConfirm: (payload: PartialRegenerationPayload) => void

onClose: () => void

}
```

Supporting Types

```
type PartialRegenerationPayload = {

  scope: "class" | "teacher" | "day" | "subject"

  classId?: string

  teacherId?: string

  weekdayNumber?: number

  subjectId?: string

}
```

State

```
type PartialRegenerationModalState = {

  scope: "class" | "teacher" | "day" | "subject"

  classId: string | null

  teacherId: string | null

  weekdayNumber: number | null

  subjectId: string | null

}
```

Actions

- change scope
- select target entity

- submit regeneration

25) Publish Draft Modal

Component Name

PublishTimetableVersionModal

Responsibility

Draft. تأكيد نشر نسخة

Props

```
type PublishTimetableVersionModalProps = {  
  
  open: boolean  
  
  version: TimetableVersionCardItem | null  
  
  blockingIssuesCount: number  
  
  submitting: boolean  
  
  onConfirm: () => void  
  
  onClose: () => void  
  
}
```

State

إذا احتجت تأكيد إضافي confirmedRisk: boolean قد يحتوي على

Actions

- publish
- cancel

26) Archive Version Modal

Component Name

ArchiveTimetableVersionModal

Responsibility

تأكيد أرشفة نسخة.

Props

```
type ArchiveTimetableVersionModalProps = {  
  open: boolean  
  
  version: TimetableVersionCardItem | null  
  
  submitting: boolean  
  
  onConfirm: (reason?: string) => void  
  
  onClose: () => void  
  
}
```

State

```
type ArchiveTimetableVersionModalState = {  
  reason: string  
  
}
```

27) Session Details Drawer

Component Name

TimetableSessionDetailsDrawer

Responsibility

عرض تفاصيل حصة واحدة.

Props

```
type TimetableSessionDetailsDrawerProps = {  
  open: boolean  
  
  session: TimetableSessionDetails | null  
  
  loading: boolean  
  
  onClose: () => void  
  
  onLock: (entryId: string) => void
```

```
onUnlock: (entryId: string) => void

onRegenerateRelated: (entryId: string) => void

onViewDiagnostics: (entryId: string) => void

}
```

Supporting Types

```
type TimetableSessionDetails = {

  id: string

  subjectName: string

  teacherName: string

  className: string

  gradeName: string

  weekdayLabel: string

  startTime: string

  endTime: string

  entryType: string

  roomName?: string

  isAiGenerated: boolean

  isLocked: boolean

  status: "scheduled" | "modified" | "locked" | "cancelled"

  notes?: string

  warnings: string[]

}
```

Actions

- lock entry
- unlock entry
- regenerate related area
- open diagnostics

28) Toast / Feedback Component

Component Name

TimetableFeedbackToast

Responsibility

إظهار رسائل نجاح/تحذير/فشل.

Props

```
type TimetableFeedbackToastProps = {  
  open: boolean  
  payload: {  
    type: "success" | "warning" | "error" | "info"  
    title: string  
    message: string  
  } | null  
  onClose: () => void  
}
```

29) Loading Skeleton Set

Component Name

TimetablePageSkeleton

Responsibility

أثناء تحميل الصفحة UI loading عرض.

Props

```
type TimetablePageSkeletonProps = {  
  showInsights?: boolean  
  showVersions?: boolean
```

```
}
```

30) Error State Component

Component Name

TimetablePageErrorState

Responsibility

عرض فشل تحميل الصفحة أو خطأ حرج.

Props

```
type TimetablePageErrorStateProps = {  
  message: string  
  
  onRetry: () => void  
  
  onGoBack?: () => void  
}
```

31) Suggested Data Contracts

Readiness Summary

```
type ReadinessSummary = {  
  overallStatus: "ready" | "partial" | "blocked"  
  
  totalChecks: number  
  
  passedChecks: number  
  
  failedChecks: number  
  
  warningChecks: number  
  
  items: ReadinessItem[]  
}
```

Grid Data

```
type TimetableGridData = {  
  
  versionId: string
```

```

versionStatus: "draft" | "published" | "archived"

mode: ViewMode

columns: TimetableGridColumn[]

rows: TimetableGridRow[]

entries: TimetableGridEntry[]

}

```

Option Item

```

type OptionItem = {

  label: string

  value: string

}

```

32) Event Map Between Components

Primary events

- TimetableActionBar.onGenerateClick
 - opens AITimetableGenerationModal
- AITimetableGenerationModal.onConfirm
 - calls page-level generateTimetable()
- TimetableViewControls.onVersionChange
 - updates selectedVersionId
 - reloads grid + issues + insights
- TimetableCell.onClick(entryId)
 - opens TimetableSessionDetailsDrawer
- TimetableVersionCard.onPublish()
 - opens PublishTimetableVersionModal
- PublishTimetableVersionModal.onConfirm()
 - calls publishVersion(versionId)
- TimetableIssuesSection.onFixIssue(issueId)
 - open related route or partial regeneration flow

33) Page-Level Backend Action Mapping

page container: هذه الدوال يجب أن يستهلكها الـ

async function fetchTimetablePageSummary()

async function fetchReadinessSummary()

async function fetchTimetableGrid(params)

async function fetchTimetableInsights(versionId)

async function fetchTimetableVersions()

async function fetchTimetableIssues(versionId)

async function fetchDiagnostics(versionId)

async function fetchSessionDetails(entryId)

async function generateFullTimetable(payload)

async function partiallyRegenerateTimetable(payload)

async function publishTimetableVersion(versionId)

async function archiveTimetableVersion(versionId, reason)

async function lockTimetableEntry(entryId)

async function unlockTimetableEntry(entryId)

34) Rendering Rules

Rules

- PrincipalTimetablePage يقرر ما إذا كانت الصفحة:
 - loading
 - error
 - empty
 - ready
- TimetableGridSection لا يرسم TimetableGrid إلا إذا:
 - loading = false
 - gridData != null
- TimetableActionBar يجب أن يعرف سبب تعطيل أي زر
- TimetableStatusBanner وليس على تخمينات من الواجهة current state يعتمد فقط على
- TimetableSessionDetailsDrawer لا يفتح بدون selectedSessionEntryId

35) Recommended Folder Structure

/components/timetable/

PrincipalTimetablePage.tsx

TimetablePageHeader.tsx

TimetableStatusBanner.tsx

TimetableActionBar.tsx

AltTimetableGenerateButton.tsx

TimetableReadinessPanel.tsx

ReadinessChecklistItem.tsx

TimetableViewControls.tsx

TimetableViewModeTabs.tsx

TimetableFiltersBar.tsx

TimetableSearchInput.tsx

TimetableToggleControls.tsx

TimetableGridSection.tsx

TimetableGrid.tsx

TimetableCell.tsx

TimetableEmptyGridState.tsx

TimetableInsightsPanel.tsx

InsightMetricCard.tsx

TimetableVersionManager.tsx

TimetableVersionCard.tsx

TimetableIssuesSection.tsx

TimetableDiagnosticsModal.tsx

AltTimetableGenerationModal.tsx

PartialRegenerationModal.tsx

PublishTimetableVersionModal.tsx

ArchiveTimetableVersionModal.tsx

TimetableSessionDetailsDrawer.tsx

TimetableFeedbackToast.tsx

TimetablePageSkeleton.tsx

TimetablePageErrorState.tsx

36) Final Implementation Rule for the Agent

Every UI responsibility must be isolated in a dedicated component.

The page container should orchestrate data fetching, state management, and backend actions, while child components remain focused on display logic, user interaction, and event emission.

The component system must support:

- clear data flow
- reusable controls
- draft/published timetable lifecycle
- interactive grid cells
- readiness validation
- diagnostics visibility
- full and partial generation flows

37) Short Developer Handoff Version

Build the Principal Timetable Page as a modular component system.

Root page:

- PrincipalTimetablePage

Main sections:

- TimetablePageHeader

- TimetableStatusBanner

- TimetableActionBar

- TimetableReadinessPanel

- TimetableViewControls

- TimetableGridSection

- TimetableInsightsPanel

- TimetableVersionManager

- TimetableIssuesSection

Modals:

- AllTimetableGenerationModal
- PartialRegenerationModal
- PublishTimetableVersionModal
- ArchiveTimetableVersionModal
- TimetableDiagnosticsModal
- TimetableSessionDetailsDrawer

Each component must have:

- clear responsibility
- typed props
- predictable state
- explicit actions
- proper empty/loading/error states